Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two capacitors of capacitances C_1 and C_2 have a total capacitance of $20\mu\text{F}$ when connected in parallel and a total capacitance $4.8\mu\text{F}$ when connected in series. Find C_1 and C_2.

(A) 8 and 12

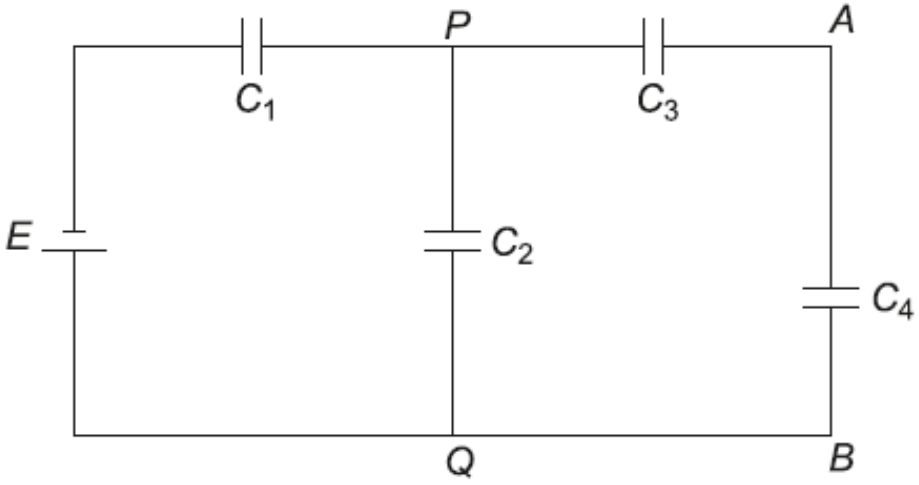
(B) 18 and 12

(C) 6 and 12

(D) 4 and 12

Q.2

The circuit shown in Fig. consists of four capacitors C_1, C_2, C_3, and C_4 each of capacitance $4\mu\text{F}$ and a battery of emf $E = 5\ \text{V}$. Find the potential difference between A and B.



(A) 1 V

(B) 2 V

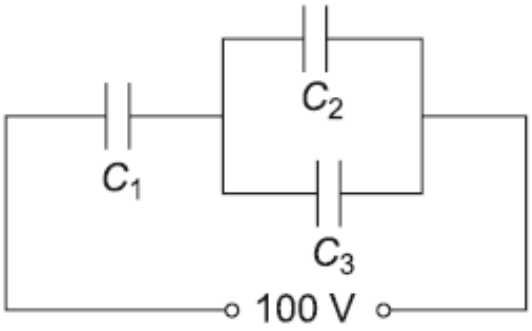
(C) 4 V

(D) 5 V

Q.3

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Find the charge on each capacitor shown in Fig. Given $C_1 = 2\mu\text{F}$, and $C_2 = 2\mu\text{F}$ and $C_3 = 1\mu\text{F}$.



(A) Q1 = 120 microcoulomb, Q2 = 80 microcoulomb and Q3 = 40 microcoulomb

(B) Q1 = 120 microcoulomb, Q2 = 180 microcoulomb and Q3 = 140 microcoulomb

(C) Q1 = 150 microcoulomb, Q2 = 80 microcoulomb and Q3 = 120 microcoulomb

(D) Q1 = 150 microcoulomb, Q2 = 80 microcoulomb and Q3 = 50 microcoulomb

Q.4

A capacitor of capacitance C_1 is charged by connecting it to a battery. The battery is now removed and this capacitor is connected to a second uncharged capacitor of capacitance C_2. If the charge distributes equally on the two capacitors, the ratio of the total energy stored in the capacitors after connection to the total energy stored in them before connection is

(A) 1

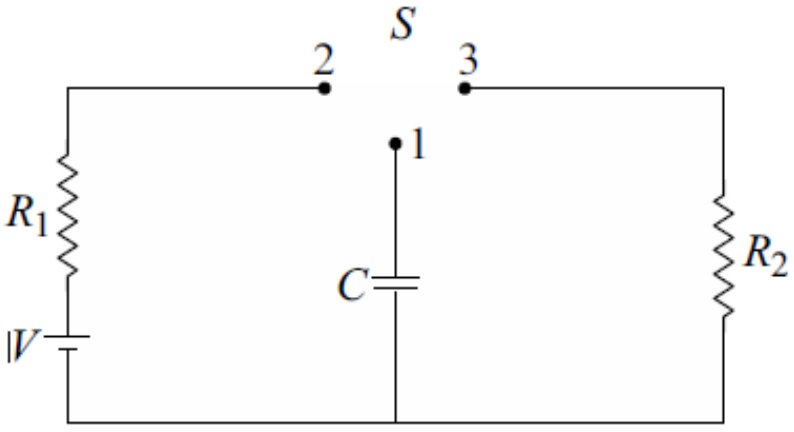
(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\frac{1}{3}$

Q.5

In fig. 1 is disconnected from 2 and connected to 3 at time $t = 0$. The time taken by the power dissipated through R_2 to decrease to one half the initial power is



(A) $\frac{R_2 C}{2}\ln(2)$

(B) $\frac{2\ln(2)}{R_2 C}$

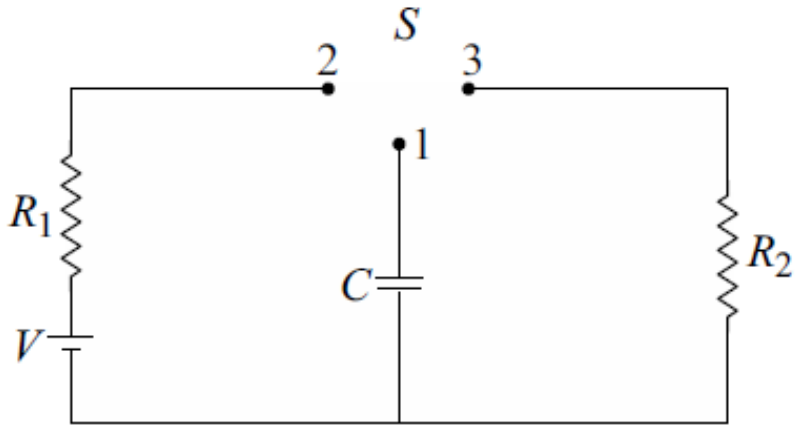
(C) $\frac{R_2 C}{2}\ln\left(\frac{1}{2}\right)$

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(D) $\frac{2\ln(1/2)}{R_2C}$

Q.6

Figure shows as uncharged capacitor of capacitance C, two resistors of resistances R_1 and R_2 a battery of voltage V and a two-way switch S. Terminal 1 of S is connected to terminal 2 . The time required for the charge on the capacitor to reach half of the final charge is



(A) $R_1C\ln(1/2)$

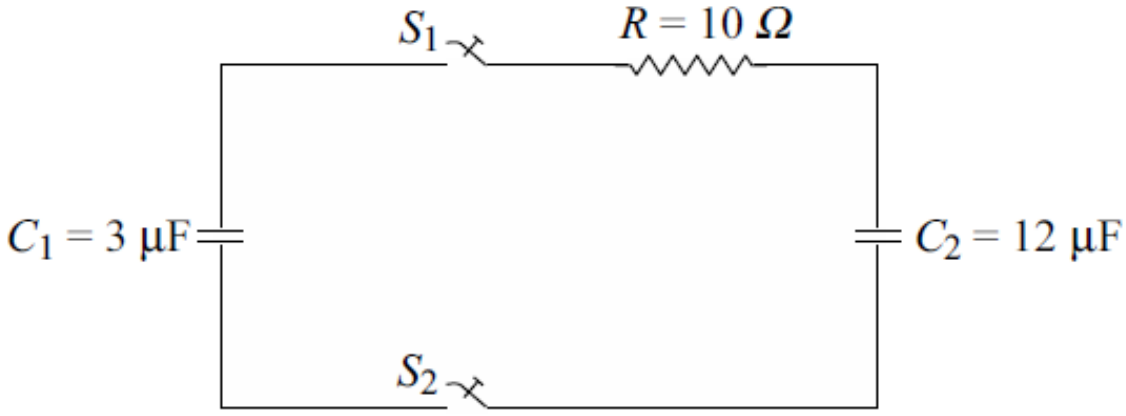
(B) $\frac{R_1C}{\ln\left(\frac{1}{2}\right)}$

(C) $R_1C\ln(2)$

(D) $\frac{R_1C}{\ln(2)}$

Q.7

Two capacitors C_1 and C_2 are connected to resistor R via two switchs S_1 and S_2 as shown in Fig. Capacitor C_1 is initially charged to a voltage $V_1 = 15$ V. Both the switchs are then closed simultaneously. When the steady state is established, the charges on C_1 and C_2 respectively are



(A) $6\mu C, 24\mu C$

(B) $9\mu C, 36\mu C$

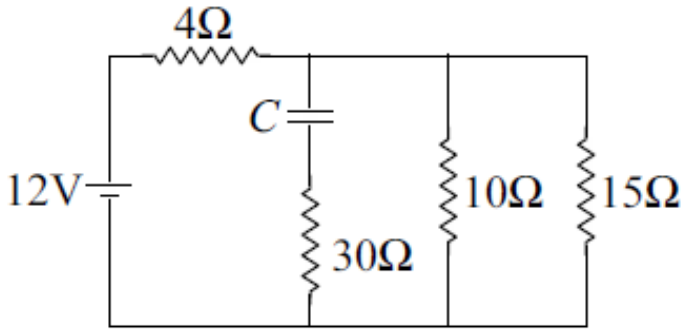
(C) $12\mu C, 48\mu C$

(D) $15\mu C, 60\mu C$

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Q.8

In the circuit shown in Fig., the capacitance of the capacitor is $C = 5\mu\text{F}$. When the steady state is reached, the charge on the capacitor is



(A) $9\mu\text{C}$

(B) $18\mu\text{C}$

(C) $24\mu\text{C}$

(D) $36\mu\text{C}$

Q.9

A capacitor of capacitance C is connected to a $6.0\ \text{V}$ battery through a resistance of 5Ω. The potential difference between the plates rises from zero to $3.0\ \text{V}$ in $6.9\mu\text{s}$. The value of C is

(A) $1\mu\text{F}$

(B) $2\mu\text{F}$

(C) $3\mu\text{F}$

(D) $4\mu\text{F}$

Q.10

Let C be the capacitance of a capacitor discharging through a resistor R. Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 2

Q.11

A parallel plate capacitor with air between the plates has a capacitance of 9pF. The separation between its plates is ' d '. The space between the plates is now filled with two dielectrics. One dielectric has dielectric

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constant $K_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $K_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now

(A) 40.5pF

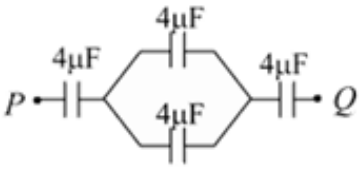
(B) 20.25pF

(C) 1.8pF

(D) 45pF

Q.12

Four condensers each of capacity $4\mu\text{F}$ are connected as shown in figure. $V_P - V_Q = 15$ volts. The energy stored in the system is



(A) 2400 ergs

(B) 1800 ergs

(C) 3600 ergs

(D) 5400 ergs

Q.13

The work done in placing a charge of 8×10^{-18} coulomb on a condenser of capacity 100 microfarad is

(A) 3.1×10^{-26} joule

(B) 4×10^{-10} joule

(C) 32×10^{-32} joule

(D) 16×10^{-32} joule

Q.14

Two identical capacitors, have the same capacitance C. One of them is charged to potential V_1 and the other to V_2. The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the
combined system is

(A) $\frac{1}{4}C\left(V_1^2 - V_2^2\right)$

(B) $\frac{1}{4}C\left(V_1^2 + V_2^2\right)$

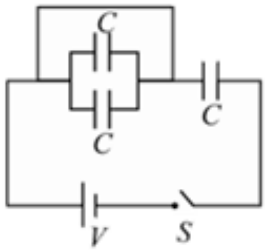
(C) $\frac{1}{4}C\left(V_1 - V_2\right)^2$

(D) $\frac{1}{4}C\left(V_1 + V_2\right)^2$

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Q.15

In the given circuit, find the heat generated if switch S is closed.



(A) $\frac{3}{2}CV^2$

(B) $\frac{1}{2}CV^2$

(C) CV^2

(D) $\frac{1}{3}CV^2$

Answers & Solutions

Q.1 Answer:

8 and 12

Solution:

$C_1 + C_2 = 20$ **(1)**

$\frac{C_1C_2}{C_2+C_2} = 4.8$ **(2)**

Using (1) in (2) we get

$\frac{C_1C_2}{20} = 4.8 \Rightarrow C_1C_2 = 96\mu\text{F}$

$\therefore C_2 = \frac{96}{C_1}$

Using this in (2) we get

$C_1^2 - 20C_1 + 96 = 0$

The two roots of this equation are $C_1 = 8\mu\text{F}$ and $12\mu\text{F}$. Hence the two capacitors have capacitances of $8\mu\text{F}$ and $12\mu\text{F}$

Q.2 Answer:

1 V

Solution:

Capacitors C_3 and C_4 are in series, their equivalent capacitance is

$C' = \frac{C_3C_4}{(C_3+C_4)} = \frac{4\times4}{(4+4)} = 2\mu\text{F}$

C' is in parallel with C_2. Their equivalent capacitance is

$C'' = C_2 + C' = 4 + 2 = 6\mu\text{F}$

Hence the circuit can be redrawn as shown in Fig.

If V_1 is the p.d. across C_1 and V'' across C'', then (since they are in series)

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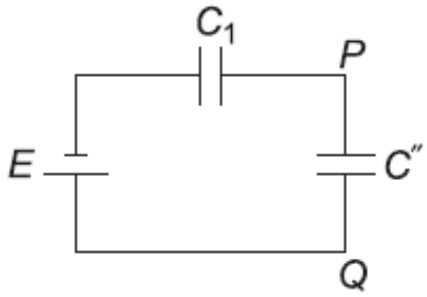
$$C_1V_1 = C''V''$$

$$\Rightarrow \quad 4V_1 = 6V'' \Rightarrow V_1 = \frac{3V''}{2}$$

But $V_1 + V'' = E = 5$. Hence

$\frac{3V''}{2} + V'' = 5 \Rightarrow V'' = 2$ V

Thus the p.d. between P **and** $Q = 2$ V. **By symmetry of** C_3 **and** C_4, **the p.d. between** A **and** $B = 1$ V.



Q.3 Answer:

Q1 = 120 microcoulomb, Q2 = 80 microcoulomb and Q3 = 40 microcoulomb

Solution:

The capacitance of the parallel combination of C_2 **and** C_3 **is** $C' = C_2 + C_3 = 3\mu$F

The circuit can be redrawn as shown in Fig.

The equivalent capacitance C **of series combination of** C_1 **and** C' **is given by**

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C'}$$

$$\Rightarrow \quad C = \frac{C_1C'}{C_1+C'} = \frac{2\times3}{2+3} = \frac{6}{5} = 1.2\mu\text{F}$$

Let Q_1, Q_2 **and** Q_3 **be the charges on** C_1, C_2, **and** C_3 **respectively. Since** C_1 **and** C' **are in series, charge on** C_1 **= charge on** $C' = Q_1$. **Let** V_1 **and** V' **be the potential differences across** C_1 **and** C' **respectively. Then**

$Q_1 = C_1V_1 = C'V' \Rightarrow \frac{V'}{V_1} = \frac{C_1}{C'} = \frac{2}{3}$

But $V_1 + V' = 100$. Therefore

$V_1 + \frac{2}{3}V_1 = 100 \Rightarrow V_1 = 60$ V

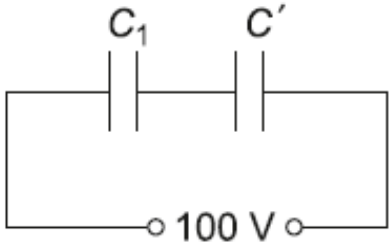
Therefore $V' = 100 - 60 = 40$ V. Hence

$Q_1 = C_1V_1 = 2\mu\text{F} \times 60\text{ V} = 120\mu\text{C}$

Since C_2 **and** C_3 **are in parallel, potential difference** V_2 **across** C_2 **= potential difference** V_3 **across** $C_3 = V' = 40$ V **Therefore, Charge on capacitor** $C_2 = C_2V_2 = 2\mu\text{F} \times 40\text{ V} = 80\mu\text{C}$

and Charge on capacitor $C_3 = C_3V_3 = 1\mu\text{F} \times 40\text{ V} = 40\mu\text{C}$

Thus $Q_1 = 120\mu\text{C}, Q_2 = 80\mu\text{C}$ **and** $Q_3 = 40\mu\text{C}$



Q.4 Answer:

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$\frac{1}{2}$

Solution:

If Q is the initial charge on capacitor C_1, the initial energy is given by

$U_i = Q^2/2C_1$

When the two capacitors are connected together, and as the charge is distributed equally, the charge on

each capacitor is $Q/2$. Since the potential difference (in a parallel connection) across the two capacitors is also the same, it follows that their capacitances are equal (since $C = Q/V$). Thus $C_1 = C_2 = C$ (say). Also, $Q_1 = Q_2 = Q/2$

Therefore, final energy stored in the two capacitors is

$U_f = \frac{Q_1^2}{2C_1} + \frac{Q_2^2}{2C_2} = \frac{(Q/2)^2}{2C} + \frac{(Q/2)^2}{2C} = \frac{Q^2}{4C}$

But $U_i = \frac{Q^2}{2C}$

$\therefore \quad \frac{U_f}{U_i} = \frac{1}{2}$, which is choice (b).

Q.5 Answer:

$\frac{R_2 C}{2}\ln(2)$

Solution:

The current due to discharging of the capacitor decreases with time t as

$I = I_0 e^{-t/\tau_2}$

where $I_0 = \frac{V}{R_2}$ is the maximum current and $\tau_2 = R_2 C$. The power dissipated through R_2 is

$P = I^2 R_2 = I_0^2 R_2 e^{-2t/\tau_2} = P_0 e^{-2t/\tau_2}$

where $P_0 = I_0^2 R_2$ is the maximum power dissipated. For P to become $P_0/2$, the time t required is given by

$$\Rightarrow \quad \frac{P_0}{2} = P_0 e^{-2t/\tau_2}$$

$$\Rightarrow \quad e^{-2t/\tau_2} = \frac{1}{2}$$

$$\Rightarrow \quad e^{2t/\tau_2} = 2$$

$$\Rightarrow \quad \frac{2t}{\tau_2} = \ln(2)$$

$$\Rightarrow t = \frac{\tau_2 \ln(2)}{2} = \frac{R_2 C \ln(2)}{2}$$

So the correct choice is (a).

Q.6 Answer:

$R_1 C \ln(2)$

Solution:

The charge on the capacitor at time t is given by

$Q = Q_0\left(1 - e^{-t/\tau_1}\right)$

where time constant $\tau_1 = R_1 C$ and Q_0 is the final charge, i.e., the charge when the capacitor is fully charged to the voltage of the battery. For Q to become equal to $\frac{Q_0}{2}$, the time t required is given

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by

$$\frac{Q_0}{2} = Q_0\left(1 - e^{-t/\tau_1}\right)$$

$$\Rightarrow 1 - e^{-t/\tau_1} = \frac{1}{2}$$

$$\Rightarrow e^{-t/\tau_1} = \tfrac{1}{2}$$

$$\Rightarrow \quad e^{-t/\tau_1} = 2$$

$$\Rightarrow \quad \tfrac{t}{\tau_1} = \ln(2)$$

$$\Rightarrow \quad t = \tau_1 \ln(2) = R_1 C \ln(2)$$

So the correct choice is (c).

Q.7 Answer:

$9\mu C, 36\mu C$

Solution:

Initial charge as C_1 is

$Q = C_1 V_1 = 3\mu\mathrm{F} \times 15\ \mathrm{V} = 45\mu\mathrm{C}.$

When S_1 and S_2 are closed, this charge will flow from C_1 to C_2 until both the capacitors have the same potential difference V. Let Q_1 and Q_2 are the charges on C_1 and C_2 when steady state is established. Since C_1 and C_2 are in parallel, V is the same in both, i.e.,

$$V = \frac{Q_1}{C_1} = \frac{Q_2}{C_2}$$

$$\text{or} \quad \frac{Q_1}{C_1} = \frac{Q - Q_1}{C_2}$$

$$\Rightarrow \quad \tfrac{Q_1}{3} = \tfrac{45 - Q_1}{12}$$

$$\Rightarrow \quad Q_1 = 9\mu\mathrm{C}$$

and $Q_2 = Q - Q_1 = 45 - 9 = 36\mu\mathrm{C}$. So the correct choice is (b).

Q.8 Answer:

$36\mu\mathrm{C}$

Solution:

In the steady state, no current flows through the branch containing the capacitor. Charge on the capacitor is $Q = CV$, where V is the potential difference between the capacitor plates [see Fig.] The 30Ω resistance carries no current and hence can be ignored. The resistance of circuit is

$$R = 4 + \tfrac{10\times 15}{10+15} = 10\Omega$$

$$\text{Current} \quad I = \tfrac{12\ \mathrm{V}}{10\Omega} = 1.2\ \mathrm{A}$$

Applying Kirchhoff's voltage law,

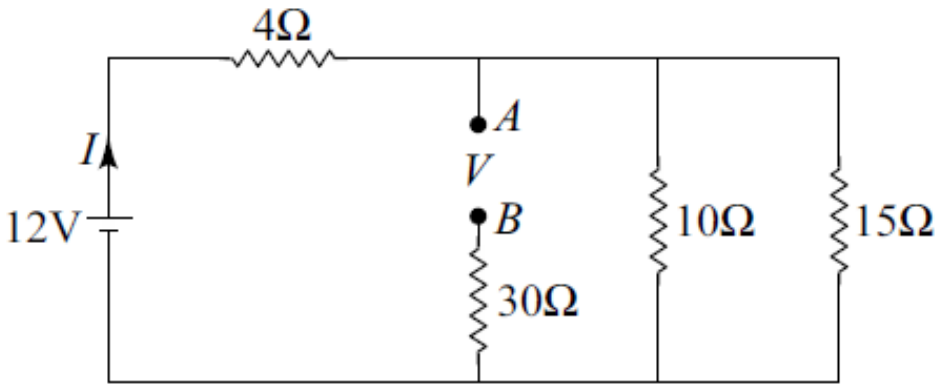
$$V_B - V_A = -12 + 4 \times 1.2 = -12 + 4.8 = -7.2\ \mathrm{V}$$

$$\therefore V = V_A - V_B = 7.2\ \mathrm{V}$$

Hence $\quad Q = CV = 5\mu\mathrm{F} \times 7.2\ \mathrm{V} = 36\mu\mathrm{C}$

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So the correct choice is (d).



Q.9 Answer:

$2\mu\text{F}$

Solution:

During charging, the charge at time t on the capacitor is given by

$Q = Q_0\left(1 - e^{-t/RC}\right)$

Therefore, the potential difference at time t between the capacitor plates is

$V = \frac{Q}{C} = \frac{Q_0}{C}\left(1 - e^{-t/RC}\right)$

$\Rightarrow V = V_0\left(1 - e^{-t/RC}\right)$

where $V_0 = \frac{Q_0}{C}$ is the final potential difference $= 6.0$ V, the voltage of the battery. Given $V = 3.0\text{ V} = \frac{V_0}{2}$. Hence

$\frac{V_0}{2} = V_0\left(1 - e^{-t/RC}\right)$

$\Rightarrow \quad 1 - e^{-t/RC} = \frac{1}{2}$

$\Rightarrow \quad e^{-t/RC} = \frac{1}{2}$

$\Rightarrow \quad \frac{t}{RC} = \ln(2)$

$\Rightarrow \quad C = \frac{t}{R\ln(2)} = \frac{6.9\mu\text{s}}{5\times 0.69} = 2\mu\text{F}$

Q.10 Answer:

$\frac{1}{4}$

Solution:

The charge decays according to the equation $Q = Q_0 e^{-t/\tau}$

where $\tau = RC$ and Q_0 = charge at $t = 0$. Energy stored is $U = \frac{Q^2}{2C} = \frac{Q_0^2}{2C}e^{-2t/\tau} = U_0 e^{-2t/\tau}$

where U_0 = initial energy at $t = 0$.

Given $U = \frac{U_0}{2}$ at $t = t_1$.

i.e. $\frac{U_0}{2} = U_0 e^{-2t_1/\tau} \Rightarrow e^{2t_1/\tau} = 2$

or $\frac{2t_1}{\tau} = \ln(2) \Rightarrow t_1 = \frac{\tau}{2}\ln(2)$

Given $Q = \frac{Q_0}{4}$ at $t = t_2$. Hence

$\frac{Q_0}{4} = Q_0 e^{-t_2/\tau}$

which gives $t_2 = \tau\ln(4) = 2\tau\ln(2)$

$\therefore \quad \frac{t_1}{t_2} = \frac{1}{4}$**, which is choice (c).**

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Q.11 Answer:

40.5pF

Solution:

Given $C_0 = \frac{\varepsilon_0 A}{d} = 9\text{pF}$

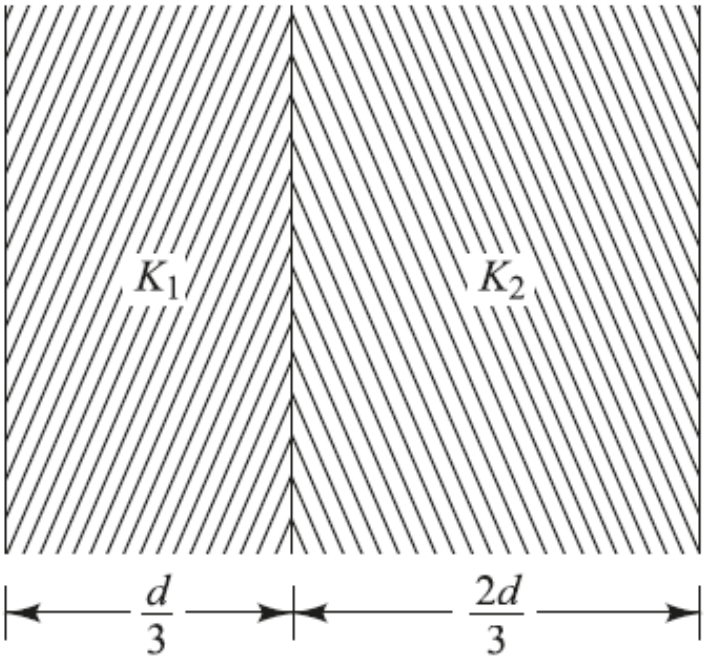
$C_1 = \frac{k_1 \varepsilon_0 A}{d/3} = \frac{3\varepsilon_0 A}{d/3} = \frac{9\varepsilon_0 A}{d} = 9C_0$

$C_2 = \frac{k_2 \varepsilon_0 A}{2d/3} = \frac{6\varepsilon_0 A}{2d/3} = \frac{9\varepsilon_0 A}{d} = 9C_0$

Capacitors C_1 **and** C_2 **are in series. The equivalent capacitance is**

$$\text{C} = \frac{C_1 C_2}{C_1 + C_2} = \frac{9C_0 \times 9C_0}{18C_0} = \frac{9}{2}C_0$$

$$= \frac{9}{2} \times 9\text{pF} = 40.5\text{pF}$$



Q.12 Answer:

1800 ergs

Solution:

Total capacitance of given system $C_{eq} = \frac{8}{5}\mu\text{F}$

$U = \frac{1}{2}C_{eq}V^2 = \frac{1}{2} \times \frac{8}{5} \times 10^{-6} \times 225 = 180 \times 10^{-6} J = 180 \times 10^{-6} \times 10^7$ **erg** $=$ 1800erg

Q.13 Answer:

32×10^{-32} joule

Solution:

$$\text{Work done } = \frac{1}{2}\frac{\text{q}^2}{\text{C}} = \frac{\left(8 \times 10^{-18}\right)^2}{2 \times 100 \times 10^{-6}}$$

$$= 32 \times 10^{-32}\ \text{J}$$

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Q.14 Answer:

$\frac{1}{4}C\left(V_1-V_2\right)^2$

Solution:

Initial energy of the system $U_i=\frac{1}{2}CV_1^2+\frac{1}{2}CV_2^2$

When the capacitors are joined, common potential

$V=\frac{CV_1+CV_2}{2C}=\frac{V_1+V_2}{2}$

Final energy of the system

$U_f=\frac{1}{2}(2C)V^2=\frac{1}{2}2C\left(\frac{V_1+V_2}{2}\right)^2=\frac{1}{4}C\left(V_1+V_2\right)^2$

Decreased in energy $=U_i-U_f=\frac{1}{4}C\left(V_1-V_2\right)^2$

Q.15 Answer:

$\frac{1}{2}CV^2$

Solution:

$C_{\text{eq}}=C$

So, work done by battery $=CV^2$

Heat generated $=\frac{1}{2}CV^2$

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